# Customer Churn Risk Report

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## Prediction

## **Likely to stay** - churn probability **14.3%**

| | |
|---|---|
| Risk level | Low |
| Decision threshold | 33.0% (above this = likely to churn) |
| Retention probability | 85.7% |

The model estimates a low risk of leaving. No urgent action is indicated.

## Customer details

| | |
|---|---|
| Contract type | One year |
| Payment method | Electronic check |
| Internet service | No |
| Online security | Yes |
| Tech support | Yes |
| Tenure | 12 months |
| Monthly charges | $90.00 |
| Total charges | $840.00 |

## Key factors

| Input | Value | Effect on estimate |
|---|---|---|
| Tenure | 12 months | Pushes the estimate up |
| Contract type | One year | Pushes the estimate down |
| Online security | Yes | Pushes the estimate down |
| Payment method | Electronic check | Pushes the estimate up |
| Tech support | Yes | Pushes the estimate down |

Factors describe patterns the model associates with churn, not proven causes.

## Recommended retention actions

• Invite the customer to an automatic payment method (bank transfer or card).

## Model information

| | |
|---|---|
| Model | Gradient Boosting |
| Evaluation | 1409 held-out customers (stratified 20% test split, seed 42) |
| Accuracy / precision / recall | 76.1% / 53.6% / 75.1% |
| F1-score | 62.6% |
| ROC-AUC | 0.838 |
| PR-AUC (baseline = churn rate) | 0.657 (baseline 0.265) |
| Brier score (lower is better) | 0.138 (baseline 0.195) |
| Confusion matrix (TN/FP/FN/TP) | 792 / 243 / 93 / 281 |

| | |
|---|---|
| Data | Telco-Customer-Churn.csv (7,043 customers) |

## How to read the probability

• The predicted outcome compares the estimated probability with the decision threshold.

• Risk bands: Low below 40%, Moderate from 40% to below 70%, High from 70%.

• Key factors list the inputs that moved this customer's estimate the most.

## Limitations

• The model was trained on one public telecom dataset (IBM Telco Customer Churn). Results may not transfer to other businesses, regions or time periods.

• The model does not see support history, usage, contract dates or competitor offers, which often matter for churn.

• Key factors show patterns the model associates with churn. They are not proven causes.

• Probabilities are estimates. Compare customers with each other rather than reading any single score as an exact frequency.

*This report is generated by a statistical model trained on a public telecom dataset. The churn probability is an estimate of how similar customers behaved in that data. It is not a guarantee of what this customer will do, and it does not prove why they would leave. Use it together with your own knowledge of the customer.*